Is Zero Tolerance Always Optimal? A Compact ABM Specification of Whistleblowing, Enforcement, and Cultural Feedback

Executive Summary

Employees occasionally face opportunities for misconduct and decide probabilistically based on private benefit, moral cost, and perceived detection (shaped by trust, backlog, and recent sanctions). Misconduct can be observed by coworkers, who decide whether to report depending on trust, reporting norms, fear, and expected retaliation. Reports enter a compliance queue with finite capacity; investigations yield substantiation with probability increasing in evidence quality and decreasing with backlog. Retaliation is more likely for powerful wrongdoers and less likely when anonymity and protection are strong. Department trust and norms update from visible sanctions, retaliation, and delays, feeding back into future misconduct and reporting. The model tests when stronger enforcement reduces true misconduct versus when it suppresses reporting and increases hidden misconduct.

1. Research question and outcome definition

RQ. Does strict zero-tolerance enforcement always minimize long-run misconduct, or can intermediate enforcement produce better outcomes due to fear–trust feedback?

Emergent tolerance. Effective Organizational Tolerance (EOT):

$$EOT(t) = \frac{M(t) - Sanc(t)}{M(t)} \quad \text{(share of true misconduct not sanctioned).}$$

We evaluate regimes by true misconduct $M(t)$, sanctioned misconduct $Sanc(t)$, reporting stability, retaliation, and backlog/delay.

2. Research relevance

Whistleblowing is commonly treated as a process with stages (wrongdoing → observation → reporting → organizational response), motivating explicit modeling of observation, reporting, and response rather than a single "report" decision.[1] Meta-analytic evidence highlights retaliation risk as a key determinant and shows that correlates can differ for intentions versus actions.[2] Practice evidence indicates that tips are a dominant detection channel, supporting "reporting drives detection" as a realistic mechanism.[3] Evidence from compliance practice and archival work suggests anonymous reports can be harder to substantiate, motivating an anonymity–evidence-quality tradeoff.[4]

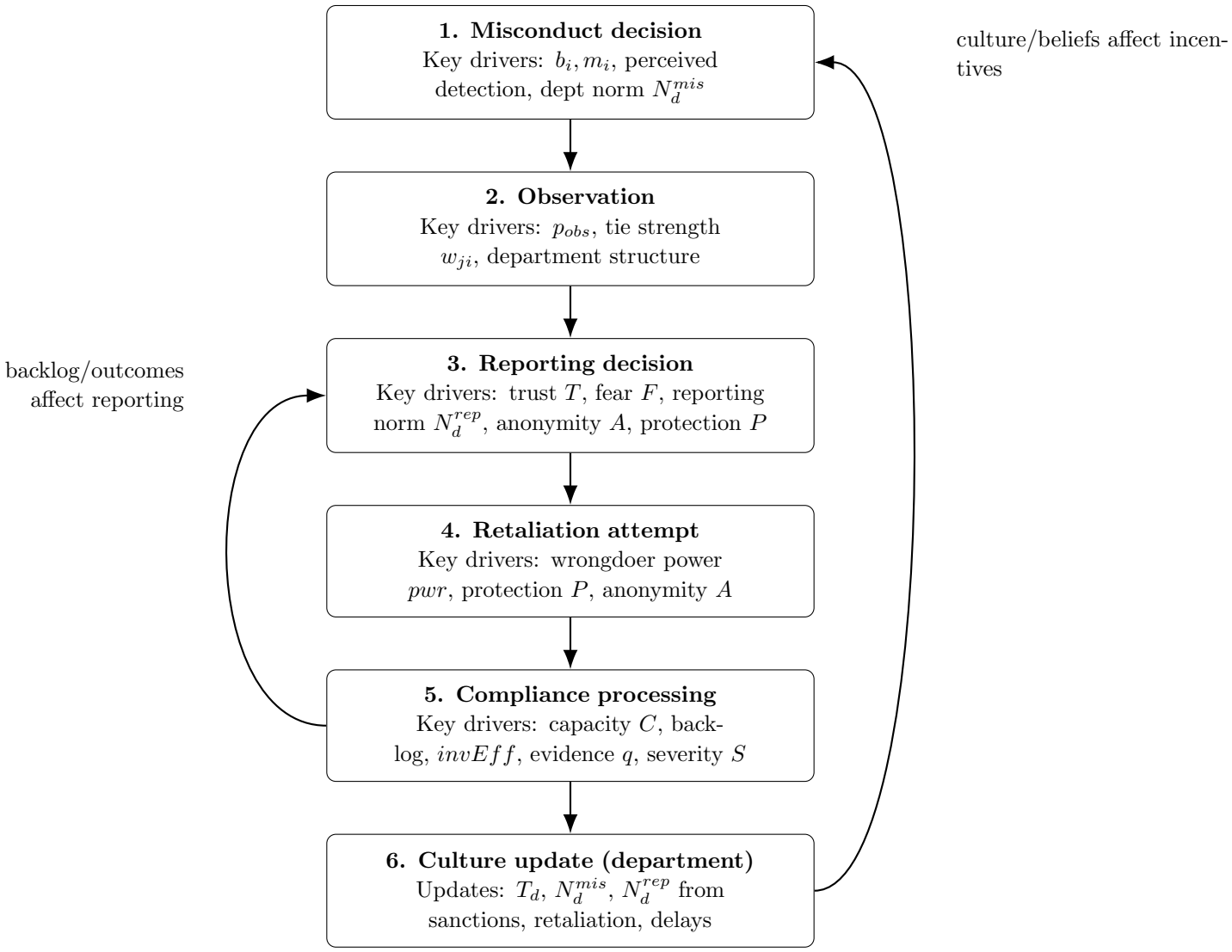
[1] Oxford Research Encyclopedia overview of whistleblowing describing Near & Miceli's process model.

[2] Mesmer-Magnus & Viswesvaran (2005), *Journal of Business Ethics*: meta-analysis (26 samples; $N = 18{,}781$) on correlates of intentions/actions and retaliation.

[3] ACFE (2024) Report to the Nations: 43% of occupational frauds detected by a tip.

[4] Bishop et al. (2021), *Journal of Forensic Accounting Research*: discusses anonymous reporting and notes lower substantiation rates in practice (e.g., NAVEX).

3. Process diagram



4. Model structure (agents, organization, information)

The organization contains N employees distributed across D departments. Employees are connected by a social network that is typically denser within departments than across them. In each simulated time step (tick), employees may commit misconduct, coworkers may observe and report, compliance investigates subject to capacity, and the outcomes feed back into departmental culture.

4.1 Employee agents

Employees are the primary agents. Each employee i is defined by a stable profile and a small set of evolving beliefs.

Stable traits:

$$b_i, \quad m_i, \quad r_i, \quad pwr_i, \quad d(i)$$

- b_i – **Benefit / temptation.** Potential private gain from misconduct; higher values increase wrongdoing incentives.
- m_i – **Moral cost.** Internal ethical disutility; higher values reduce wrongdoing.
- r_i – **Risk sensitivity.** Strength of response to retaliation risk; higher values suppress reporting when threat is high.
- pwr_i – **Hierarchical power.** Organizational influence; higher values increase retaliation ability and reduce perceived vulnerability.
- $d(i)$ – **Department membership.** Determines local interactions and which culture variables the agent is exposed to.

Evolving beliefs:

$$T_i(t), \qquad F_i(t)$$

- $T_i(t)$ – **Trust in compliance.** Belief that reporting will be handled fairly and effectively.
- $F_i(t)$ – **Fear of retaliation.** Perceived personal risk from reporting or being identified.

4.2 Department culture

Each department d stores:

$$T_d(t), \qquad N_d^{mis}(t), \qquad N_d^{rep}(t)$$

- T_d – **Department trust.** Local collective confidence in compliance and management response.
- N_d^{mis} – **Misconduct norm.** Perceived prevalence/acceptability of wrongdoing in the department.
- N_d^{rep} – **Reporting norm.** Perceived expectation that employees report observed wrongdoing.

4.3 Compliance unit

Policy parameters:

$$C, \quad invEff, \quad S, \quad P, \quad A, \quad T_{\text{timeout}}$$

- C – **Investigation capacity.** Maximum number of cases processed per tick.
- $invEff$ – **Investigation effectiveness.** Accuracy/quality of investigations and substantiation.
- S – **Sanction severity.** Magnitude of punishment when misconduct is substantiated.
- P – **Protection strength.** Effectiveness of anti-retaliation measures and shielding of reporters.
- A – **Anonymity probability.** Likelihood reports are anonymous; reduces retaliation risk but can reduce evidence quality.
- $T_{\textbf{timeout}}$ – **Delay threshold.** Maximum acceptable investigation delay before trust deteriorates.

5. Decision rules

A logistic function $\sigma(\cdot)$ converts incentives into probabilities.

Misconduct

Employees may commit misconduct depending on benefit, moral cost, department norm, and perceived detection.

Observation

Coworkers observe misconduct with probability $p_{obs} w_{ji}$.

Reporting

Observed misconduct may be reported depending on trust, fear, reporting norms, and retaliation risk.

Retaliation

If the reporter is identified, the wrongdoer may retaliate, increasing fear and reducing trust.

Compliance processing

Reports enter a queue. Up to C cases are processed per tick.

Culture update

Department trust and norms update based on visible sanctions, retaliation events, and delays.

6. Outcomes

$$M(t) = \sum_i \mathbb{I}\{M_i(t) = 1\}, \qquad Sanc(t) = \#\text{sanctions}, \qquad Rep(t) = \#\text{reports}$$

Key metrics include true misconduct prevalence, retaliation rate, backlog, and EOT.

7. Variable table

Symbol	Meaning
N, D	number of employees and departments
b_i, m_i, r_i, pwr_i	benefit, moral cost, risk sensitivity, power
T_i, F_i	individual trust and fear
$T_d, N_d^{mis}, N_d^{rep}$	department trust and norms
$C, invEff, S, P, A$	compliance parameters
q, w_{ji}	evidence quality, tie strength